



Long-term senolytic therapy with Dasatinib and Quercetin alleviates lipofuscin-dependent retinal degeneration in mice

Bo Yang^{a,b}, Kunhuan Yang^{a,b}, Ruitong Xi^{a,b}, Shiyong Li^a, Jingmeng Chen^c,
Yalin Wu^{a,b,d,*}

^a Department of Ophthalmology, The First Affiliated Hospital of Xiamen University, School of Medicine, Xiamen University, Xiamen, Fujian, China

^b Fujian Provincial Key Laboratory of Ophthalmology and Visual Science, Fujian Engineering and Research Center of Eye Regenerative Medicine, Eye Institute of Xiamen University, School of Medicine, Xiamen University, Xiamen, Fujian, China

^c School of Medicine, Xiamen University, Xiamen, Fujian, China

^d Shenzhen Research Institute of Xiamen University, Shenzhen, Guangdong, China

ABSTRACT

Dry age-related macular degeneration (AMD) is one of the common blinding eye diseases, with pathological hallmarks of lipofuscin accumulation, neuroretina atrophy and retinal pigment epithelium (RPE) degeneration. Currently, there are no effective interventions for dry AMD. Although there is already evidence suggesting a link between cellular senescence and age-related diseases, it is still unclear whether long-term senolytic therapy with Dasatinib and Quercetin (D + Q) can slow the progression of dry AMD and ultimately prevent retinal structural damage and function loss. Mice lacking the *Abca4* and *Rdh8* genes (*Abca4*^{-/-}*Rdh8*^{-/-} mice) are a preclinical model of dry AMD. In this study, we performed a 4-month senolytic therapy with D + Q on 4-month-old *Abca4*^{-/-}*Rdh8*^{-/-} mice. *Abca4*^{-/-}*Rdh8*^{-/-} mice at the age of 8 months showed obvious retinal degeneration, along with RPE senescence, lysosomal alkalization, lipofuscin accumulation and oxidative stress. Importantly, the long-term D + Q regimen significantly alleviated the degeneration of retinal structures and function in 8-month-old *Abca4*^{-/-}*Rdh8*^{-/-} mice, and it effectively repressed cellular senescence, lysosomal alkalization, lipofuscin accumulation and oxidative stress in the RPE. This study is the first to demonstrate the effect of long-term intervention with senolytics D + Q on dry AMD. Overall, these findings highlight the potential of long-term senolytic treatment as an intervention for dry AMD.

1. Introduction

Dry age-related macular degeneration (AMD) is a progressive retinal degenerative disease that presents with blurred vision and loss of central vision [1,2]. The pathological hallmarks of dry AMD are neuroretina atrophy and retinal pigment epithelium (RPE) degeneration [2,3], resulting from the deposition of lipofuscin, byproducts of the visual cycle, in the RPE with age [4,5]. ABC transporter 4 (ABCA4) and all-*trans*-retinol dehydrogenase 8 (RDH8) are pivotal proteins in the visual cycle, and their dysfunction disrupts this cycle and exacerbates lipofuscin accumulation [6–8]. There is already evidence that ABCA4 and RDH8 are risk factors for dry AMD [9,10]. Over time, mice with genetic deletions of *Abca4* and *Rdh8* (*Abca4*^{-/-}*Rdh8*^{-/-} mice) indeed exhibit primary characteristics of dry AMD, including the deterioration of neuroretina and RPE, as well as the accumulation of RPE lipofuscin [7, 11]. Currently, there are no effective treatments for dry AMD.

Cellular senescence has been implicated in the pathogenesis of many age-related diseases, including osteoarthritis and Alzheimer's disease

[12–14]. Senescent cells disrupt tissue homeostasis through the activation of senescence-associated β -galactosidase (SA- β -Gal), mammalian target of rapamycin (mTOR) signaling and cell cycle inhibitors, such as p16 and p21 [15–17]. In addition, these cells exhibit a senescence-associated secretory phenotype (SASP), characterized by the secretion of proinflammatory cytokines and matrix metalloproteases, which not only exacerbate local tissue damage but also trigger senescence in neighboring cells [18–20]. Cellular senescence affects the function of lysosomal vacuolar H⁺-ATPases (V-ATPases), leading to lysosomal alkalization [21,22]. It is worth noting that the accumulation of lipofuscin is closely associated with the dysfunction of RPE lysosomes [23,24].

The combination of dasatinib (D) and quercetin (Q) is one of the widely utilized senolytic therapies [25,26]. Dasatinib, an FDA-approved leukemia treatment, effectively induces apoptosis in senescent cells [27]. By contrast, quercetin, a plant-derived flavonoid, shows potent anti-inflammatory and antioxidant properties [28]. A previous study reveals that D + Q treatment reduces SA- β -Gal activity and clears

* Corresponding author. Department of Ophthalmology, The First Affiliated Hospital of Xiamen University, School of Medicine, Xiamen University, Xiamen, Fujian, 361003, China.

E-mail address: yalinw@xmu.edu.cn (Y. Wu).

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senescent cells at fracture sites, thereby accelerating fracture healing in aged mice [29]. Also, D + Q therapy can eliminate senescent cells in the lungs, leading to substantial improvements in the physiological function of individuals with idiopathic pulmonary fibrosis [30]. Although cellular senescence in the retina has been identified as a contributing factor to dry AMD [31,32], its relationship to lysosomal alkalization, lipofuscin accumulation and oxidative stress in the RPE remains ambiguous. Our study aims to clarify the potential that D + Q regimen inhibits RPE senescence and rescues retinal degeneration through alleviating lysosomal alkalization, lipofuscin accumulation and oxidative stress in the RPE.

2. Materials and methods

2.1. Reagents

The detailed reagents were listed in Table S1.

2.2. Animals

All animal protocols were approved by the Committee on the Ethics of Xiamen University Laboratory Animal Center (Approval number: XMULAC20200072). C57BL/6J wild-type mice were obtained from Gempharmatech Co., Ltd. *Abca4*^{-/-}*Rdh8*^{-/-} mice, also on a C57BL/6J genetic background, were generated in our previous studies [33]. Mice were fed a standard laboratory chow diet and sterile water and housed at room temperature (20–22 °C) with 55 ± 10 % humidity and a 12 h light/12 h dark schedule under specific pathogen-free conditions. 4-month-old C57BL/6J and *Abca4*^{-/-}*Rdh8*^{-/-} mice were housed normally until 8 months of age and then examined together with control groups of 4-month-old C57BL/6J and *Abca4*^{-/-}*Rdh8*^{-/-} mice. For the long-term D + Q regimen, 4-month-old *Abca4*^{-/-}*Rdh8*^{-/-} mice were randomly divided into 2 groups and given vehicle (30 % PEG400) or 5 mg/kg dasatinib plus 50 mg/kg quercetin by gavage on three consecutive days every two weeks until they were 8 months old.

2.3. SA-β-gal staining

RPE/choroids flat mounts were prepared according to our previous experimental method [34]. CellEvent™ Senescence Green Assay was used to fluorescently label SA-β-gal in mouse RPE. The RPE/choroids flat mounts were incubated with staining solution at 37 °C overnight and protected from light exposure. After incubation, the RPE/choroids were washed three times with phosphate buffer saline (PBS), and nuclei were stained with DAPI. Fluorescence images were captured using a LSM 880 confocal microscope (Zeiss, Germany).

2.4. Western blotting

RPE/choroid tissues were lysed in RIPA lysis buffer supplemented with 2 % protease inhibitors, and the protein concentrations were quantified using a BCA assay. Equal amounts of protein were separated by SDS-PAGE, then transferred to a PVDF membrane and blocked with 5 % skim milk. After the membrane was incubated with primary antibodies (1:1000 dilution) overnight at 4 °C and secondary antibodies (1:2000 dilution) for 2 h at room temperature, it was imaged on a scanner (Bio-Rad, Canada) with ECL chemiluminescence imaging. The detailed information of antibodies used for western blotting was shown in Table S1. The original images of unprocessed gels were provided in Fig. S1, with the red boxes highlighting the selected Western blot results.

2.5. Quantitative real-time PCR (qRT-PCR)

Total RNA was extracted from RPE/choroid tissues using total RNA extraction reagent. The extracted total RNA was used as a template to obtain cDNA through reverse transcription. qRT-PCR was performed in a

LightCycler 96 system (Roche, Basel, Switzerland) using the cDNA as a template. The sequences of the primers were listed in Table S2.

2.6. Detection of RPE tight junctions

RPE/choroid flat mounts were obtained by dissecting the eyeballs as we described previously [34]. The flat mounts were permeabilized with 1 % Triton X-100 for 1 h, followed by blocking in 5 % bovine serum albumin (BSA) and 0.3 % Triton X-100 for 1 h. They were then incubated with an anti-ZO-1 primary antibody (1:100 dilution) overnight at 4 °C and with an Alexa Fluor 488-conjugated secondary antibody (1:200 dilution) for 2 h at room temperature. The samples were imaged using the LSM 880 confocal microscope.

2.7. Detection of RPE lipofuscin autofluorescence

The nuclei of the RPE/choroid flat mounts were stained with DAPI, followed by imaging using the LSM 880 confocal microscope. All operations were carried out under light-protected conditions.

2.8. Fundus imaging and optical coherence tomography (OCT)

Mice were tested after being anesthetized with pentobarbital sodium and undergoing mydriasis with tropicamide. The structure and imaging of the fundus were evaluated in vivo using a retinal imaging system and an OCT system (Optoprobe; OPIMG-L, UK). Carbomer solution was used to keep the eyes moist during imaging.

2.9. Electroretinography (ERG)

ERG was conducted using an Animal Electroretinogram System (Diagnosys LLC, USA). After 24 h of dark adaptation, the mice were anesthetized by inhalation of isoflurane, and their pupils were dilated with 1 % tropicamide. Scotopic ERG responses were elicited using single-flash stimuli of varying intensities (0.01, 0.1, 1, and 10 cd s/m²) under dark-adapted conditions, while photopic ERG responses were elicited using varying intensities (1, 3, and 10 cd s/m²) under a bright-adapted background. The amplitudes of the a- and b-waves were analyzed in response to brief flashes. The c-wave was recorded under scotopic conditions using a single-flash stimulus at 30 cd s/m². To ensure consistency, all ERG recordings were performed at the same time of day.

2.10. Hematoxylin and Eosin (H&E) staining

Mouse eyeballs were embedded in paraffin and cut into 5 μm sections for H&E staining. The retinal sections were stained with H&E using standard protocols. Photographs were taken using a DM2500 microscope (Leica, Germany). The thickness of the photoreceptor outer nuclear layer (ONL) was measured at 250 μm intervals, both temporally and nasally, from the optic disc.

2.11. Transmission electron microscopy (TEM)

Mouse eyeballs were fixed in 3 % glutaraldehyde for 30 min, and then the posterior eyecups containing the RPE/choroids were carefully collected. Subsequently, the posterior eyecups were fixed overnight at 4 °C in 3 % glutaraldehyde, embedded in Spurr's resin, and sectioned into ultrathin slices. Images were acquired using an HT-7800 transmission electron microscope (Hitachi, Tokyo, Japan).

2.12. Gene expression omnibus (GEO) database analysis

The GSE29801 dataset from the GEO database was extracted and quality controlled using the R programming language (Version 4.4.1) on RStudio software (Version 2023.06.0). The gene expression data of RPE

for the geographic atrophy group (GSM738541, GSM738560, GSM738578, GSM738591, GSM738599) and the age-matched control group (GSM738488, GSM738503, GSM738518, GSM738523, GSM738527) were sorted out, and the detailed information was shown in Table S3. Based on the gene expression data, the limma package (Version 3.60.6) was used to draw a volcano map of differential gene expression. The pheatmap package (Version 1.0.12) was used to create heatmaps of senescence-related genes and lysosomal V-ATPase-related genes. The gene expression data were enriched for the senescence-related gene set (SAL_SEN_MAYO and DEMAGALHAES_AGING_UP), lysosome-related gene set (LYSOSOMAL_LUMEN_ACIDIFICATION, REGULATION_OF_LYSOSOMAL_LUMEN_PH and PROTON_TRANSPORTING_V_TYPE_ATPASE_COMPLEX) and oxidative stress-related gene set (CELL_REDOX_HOMEOSTASIS and OXIDATIVE_DAMAGE_RESPONSE) using the software of gene-set enrichment analysis (GSEA) (Version 4.3.3). According to the GSEA enrichment results, the GseaVis package (Version 0.1.0) was used to generate the GSEA enrichment map. These data were reviewed and approved by the institutional review boards at St Louis University, the University of Iowa, the University of Utah, and the University of California, Santa Barbara, and conforms to the tenets of the Declaration of Helsinki [35]. All participants or their relatives signed written informed consent.

2.13. Detection of lysosomal pH

LysoSensor™ Yellow/Blue DND-160 (PDMPO) was used to assess lysosomal pH of RPE cells in mice [36]. Eyeballs were dissected in PBS to isolate posterior eyecups containing the RPE. The posterior eyecups were incubated with 0.25 % trypsin at 37 °C and 5 % CO₂ for 25 min. Subsequently, trypsinization was terminated, and RPE cells were collected. Then the cells were centrifuged, suspended in DMEM/F12 medium containing 1 μM PDMPO, and incubated at 37 °C in 5 % CO₂ for 30 min. After three washes with PBS, the cells were resuspended in DMEM/F12 medium, and subjected to a Varioskan Flash (ThermoFisher Scientific, USA) for measuring fluorescence at 340 nm and 380 nm. The fluorescence ratio (340 nm/380 nm) was calculated for each sample to assess lysosomal pH.

2.14. Detection of total antioxidant capacity

The Total Antioxidant Capacity Assay Kit was used to evaluate total antioxidant capacity [37]. RPE/choroid tissues were homogenized in PBS to fully disrupt the tissue and release antioxidants. The mixture was then centrifuged at 12,000 rpm for 5 min at 4 °C, and the supernatant was used for assessing total antioxidant capacity using the kit's standard curve.

2.15. Detection of glutathione (GSH) levels and GSH/oxidized glutathione disulfide (GSSG) ratio

GSH levels and GSH/GSSG ratio were determined using a GSH and GSSG assay kit [37]. RPE/choroid tissues were frozen in liquid nitrogen and ground into powder. Subsequently, the samples were thoroughly vortexed with the protein remover. After incubation at 4 °C for 10 min, the samples were centrifuged at 12,000 rpm for 10 min at 4 °C, and the supernatant was used to determine the total GSH content. Some of the prepared samples followed the kit instructions to remove GSH for GSSH detection. Quantitative analysis of total GSH and GSSG was performed using the kit's standard curve. GSH levels were calculated based on the measured levels of total GSH and GSSG.

2.16. Detection of protein carbonyls

The Protein Carbonyl Detection Kit was used to detect protein carbonyl groups generated by oxidation [38]. In the protein carbonyl assay protocol, RPE/choroid tissues were homogenized, and proteins

were extracted. These proteins were then derivatized with DNPH (2, 4-dinitrophenylhydrazine) or a control solution for 15 min, followed by neutralization. The samples were subsequently loaded onto SDS-PAGE gels, and the DNP moieties were detected using an anti-DNP antibody and western blotting.

2.17. Statistical analysis

Results were expressed as the mean ± standard deviation (SD) from at least three independent experiments. Statistical analyses were conducted using one-way or two-way analysis of variance (ANOVA), followed by Tukey's multiple comparison tests. A *p*-value less than 0.05 was considered to indicate statistical significance. All calculations were performed using GraphPad Prism software (version 8.0).

3. Results

3.1. The RPE of dry AMD patients with geographic atrophy shows cellular senescence

Because RPE degeneration precedes photoreceptor degeneration in dry AMD patients [39,40], we focused on RPE cells and investigated their transcriptional heterogeneity. To determine whether cellular senescence contributes to geographic atrophy phenotypes in dry AMD patients, we first analyzed transcriptomic datasets from bulk RNA sequencing (RNA-seq) of the RPE from dry AMD patients with geographic atrophy. Volcano plot showing differential gene expression in RPE cells of dry AMD patients with geographic atrophy compared to age-matched controls ($\log_2FC > 1$, *P* value < 0.05) (Fig. 1A). Further analysis revealed that, as shown in the heatmap (Fig. 1B), the gene expression of SASP factors, including interleukins (ILs), chemokines, matrix metalloproteinases (MMPs), and cellular senescence markers, such as cyclin dependent kinase inhibitor 1A (CDKN1A) and cyclin dependent kinase inhibitor 2A (CDKN2A) [15], were significantly upregulated in the RPE of dry AMD patients with geographic atrophy. Moreover, gene-set enrichment analysis (GSEA) revealed a positive correlation in genes associated with cellular senescence processes in RPE cells of dry AMD patients with geographic atrophy compared to age-matched controls (Fig. 1C). Specifically, "SAL_SEN_MAYO" gene set showed a normalized enrichment score (NES) of 2.07 with a false discovery rate statistical value (FDR *q*-value) of <0.001, and "DEMAGALHAES_AGING_UP" gene set displayed an NES of 1.46 with an FDR *q*-value of 0.04 (Fig. 1C). Collectively, these data suggest that senescence-related signals are enriched in RPE cells of dry AMD patients with geographic atrophy.

3.2. *Abca4*^{-/-}*Rdh8*^{-/-} mice develop retinal functional and structural degeneration and RPE senescence with increasing age

Abca4^{-/-}*Rdh8*^{-/-} mice exhibit dry AMD phenotypes with age, such as RPE degeneration and neuroretina atrophy, and is therefore widely used to study the pathogenesis of dry AMD [11,41]. Retinal electrophysiological function was assessed by ERG in C57BL/6J and *Abca4*^{-/-}*Rdh8*^{-/-} mice at 4 and 8 months of age. Scotopic ERG is used to evaluate the function of rod cells [42]. The results showed that compared with age-matched C57BL/6J mice, the a- and b-wave amplitudes of scotopic ERG in *Abca4*^{-/-}*Rdh8*^{-/-} mice at 8 months of age were reduced, suggesting that the rod function of 8-month-old *Abca4*^{-/-}*Rdh8*^{-/-} mice was impaired (Fig. 2A and B). Photopic ERG serves to assess the function of cone cells [43]. The data indicated that the a- and b-wave amplitudes of photopic ERG in 8-month-old *Abca4*^{-/-}*Rdh8*^{-/-} mice were decreased compared to age-matched C57BL/6J mice, reflecting that the function of cone cells in *Abca4*^{-/-}*Rdh8*^{-/-} mice aged 8 months was damaged (Fig. 2C and D). Moreover, c-wave ERG is employed for detecting the function of RPE cells [44]. It was observed that the c-wave amplitude of 8-month-old

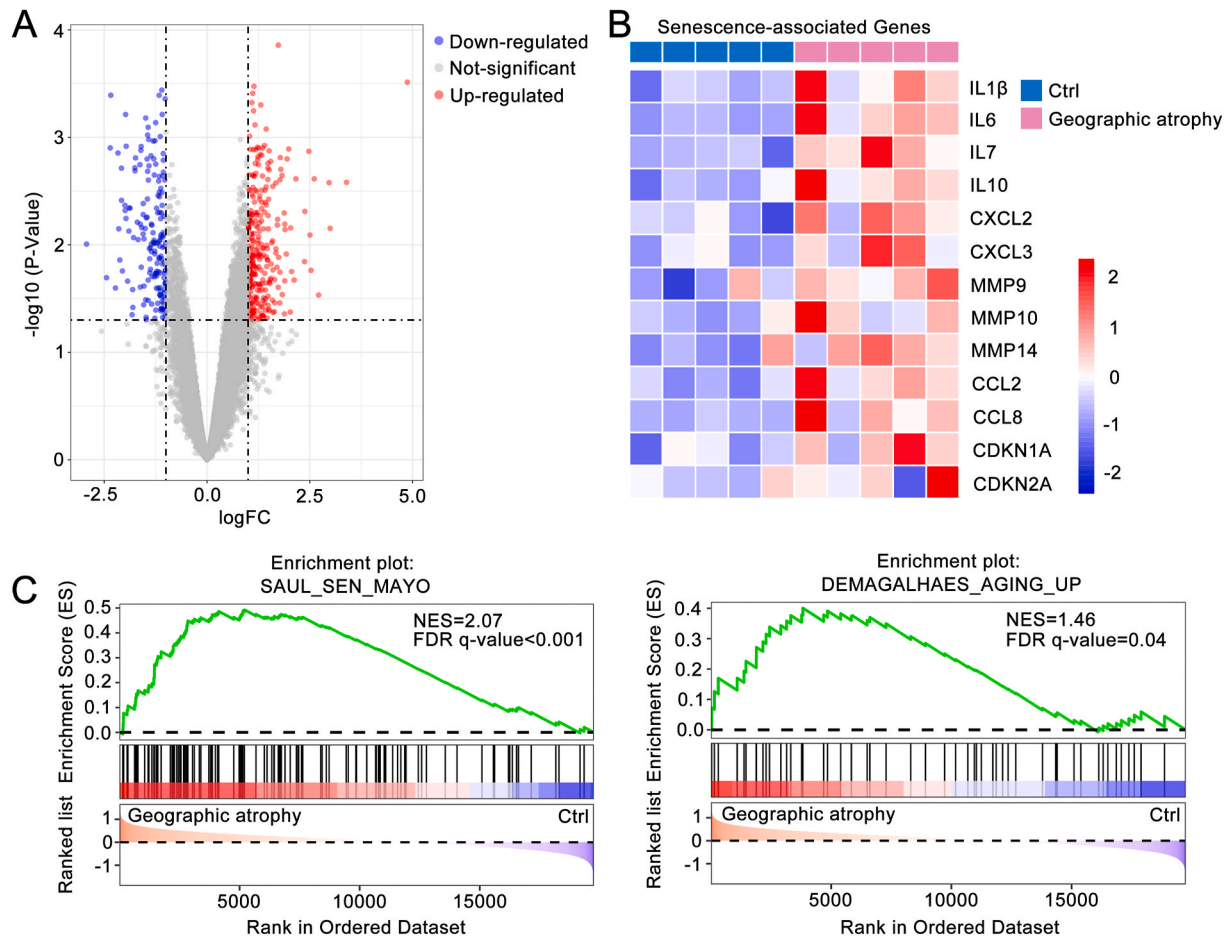


Fig. 1. Cellular senescence is observed in the RPE of dry AMD patients with geographic atrophy. **(A)** A volcano plot shows differential gene expression in RPE cells of dry AMD patients with geographic atrophy compared to controls ($\log_2\text{FC} > 1$, P value < 0.05). **(B)** A heatmap represents the profiles of cellular senescence-related genes in RPE cells of dry AMD patients with geographic atrophy compared to controls. **(C)** GSEA was performed using gene sets ‘SAUL_SEN_MAYO’ and ‘DEMAGALHAES_AGING_UP’, comparing RPE cells of dry AMD patients with geographic atrophy to normal control RPE cells.

Abca4^{-/-}*Rdh8*^{-/-} mice was visibly diminished compared with that of age-matched C57BL/6J mice, which implied that the function of RPE was significantly declined (Fig. 2E and F). By comparison with age-matched C57BL/6J mice, 8-month-old *Abca4*^{-/-}*Rdh8*^{-/-} mice exhibited distinct punctate lesions in the fundus (Fig. 2G), and showed remarkable disruption of RPE tight junctions, as revealed by ZO-1 staining (Fig. 1H). H&E staining and retinal OCT analysis disclosed that with the increase of age, mild atrophy occurred in neural retina and photoreceptor outer nuclear layer (ONL) of *Abca4*^{-/-}*Rdh8*^{-/-} mice (Fig. 2I and J). It was worth mentioning that retinal structures and function remained healthy in 8-month-old C57BL/6J and *Abca4*^{-/-}*Rdh8*^{-/-} mice compared to those in 4-month-old C57BL/6J mice. Confocal fluorescence imaging with SA- β -gal staining demonstrated that, with increasing age, cellular senescence was clearly enhanced in the RPE of *Abca4*^{-/-}*Rdh8*^{-/-} mice (Fig. 3A). The results of immunoblotting also showed that protein levels of cellular senescence markers p16, p21 and γ H2AX were dramatically increased in the RPE/choroid of 8-month-old *Abca4*^{-/-}*Rdh8*^{-/-} mice (Fig. 3B). Past studies have revealed that the mTOR/NF- κ B signaling pathway plays an important role in cellular senescence [17,45,46]. As expected, the results of western blotting indicated that phosphorylation levels of mTOR and NF- κ B were significantly elevated in the RPE/choroid of *Abca4*^{-/-}*Rdh8*^{-/-} mice aged 8 months, suggesting the activation of the mTOR/NF- κ B axis (Fig. 3C). Additionally, the results of qRT-PCR manifested that the mRNA levels of SASP-related genes, including *IL1b*, *IL6*, *MMP9*, *CCL2*, *CCL8* and *CXCL2*, were markedly upregulated in the RPE/choroid of 8-month-old *Abca4*^{-/-}*Rdh8*^{-/-} mice compared to those

in age-matched C57BL/6J mice (Fig. 3D). These findings suggest that, with the increase of age, RPE senescence is closely related to retinal degeneration in *Abca4*^{-/-}*Rdh8*^{-/-} mice.

3.3. Long-term senolytic therapy with D + Q protects retinal structures and function in *Abca4*^{-/-}*Rdh8*^{-/-} mice via decreasing RPE senescence

The long-term D + Q treatment regimen used in this study was presented in Fig. 4A. *Abca4*^{-/-}*Rdh8*^{-/-} mice aged 4 months were randomly divided into a vehicle group and a treatment group, and then received gavage treatment for 3 consecutive days every two weeks until they reached 8 months of age. Retinal electrophysiological function was assessed by ERG in *Abca4*^{-/-}*Rdh8*^{-/-} mice that underwent long-term D + Q therapy. Compared with the vehicle group, the D + Q group exhibited recovery in the a- and b-wave amplitudes of scotopic ERG and photopic ERG, indicating that long-term D + Q therapy protected the function of rods and cones (Fig. 4B–E). Notably, D + Q treatment significantly increased the c-wave amplitude in 8-month-old *Abca4*^{-/-}*Rdh8*^{-/-} mice, implying that RPE function was restored (Fig. 4F and G). Moreover, long-term D + Q therapy significantly mitigated punctate lesions in the fundus, and effectively prevented disruption of RPE tight junctions and reduction in the thickness of neural retina and photoreceptor ONL in 8-month-old *Abca4*^{-/-}*Rdh8*^{-/-} mice (Fig. 4H–K). Confocal fluorescence imaging showed that D + Q treatment visibly reduced the expression of SA- β -gal in the RPE of *Abca4*^{-/-}*Rdh8*^{-/-} mice at 8 months of age, suggesting that RPE senescence was inhibited (Fig. 5A). In the D + Q group, there was a significant reduction in the

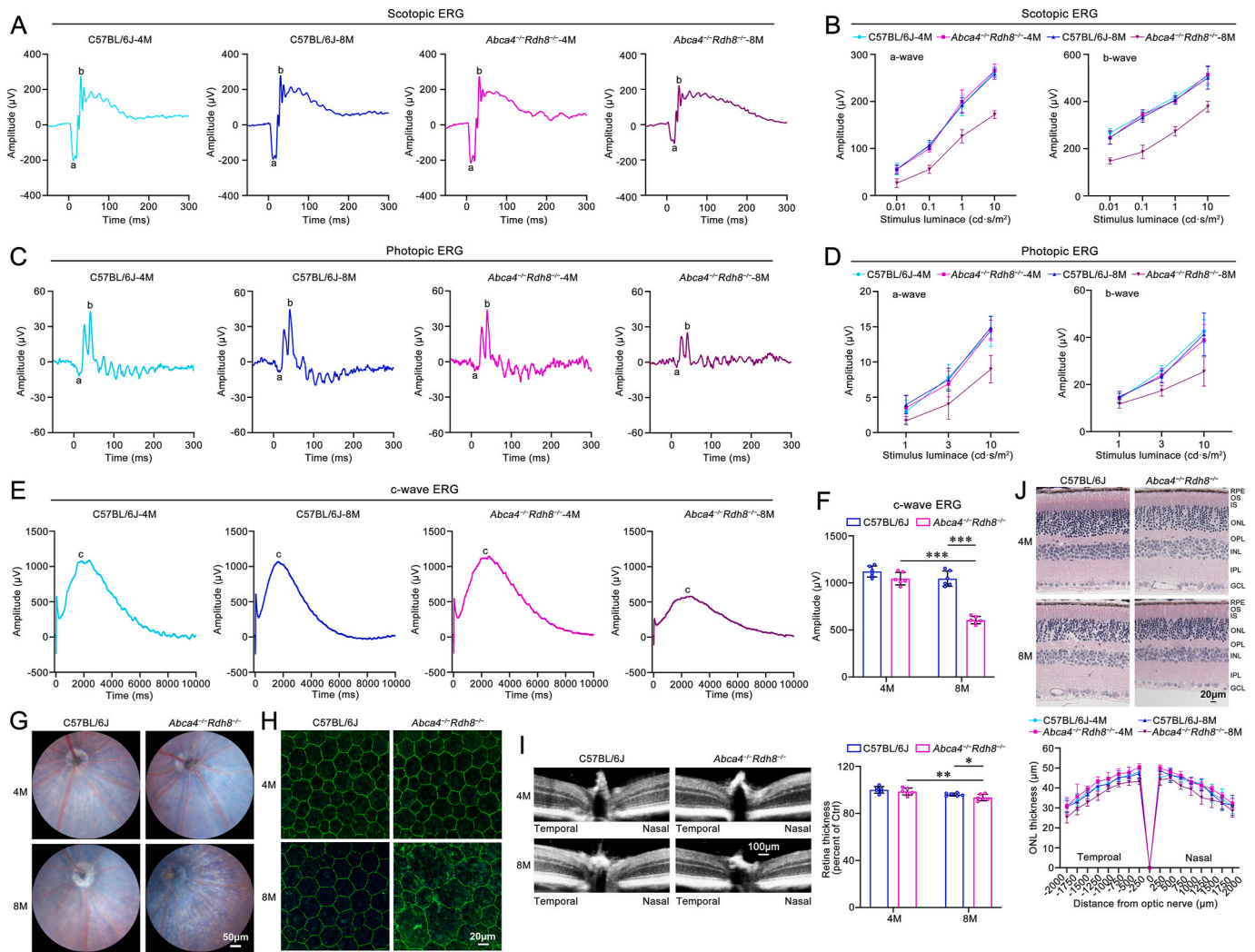


Fig. 2. *Abca4*^{-/-}*Rdh8*^{-/-} mice develop dry AMD-like retinal degeneration with increasing age. Retinal function and structures were evaluated in 4-month-old (4 M) and 8-month-old (8 M) *Abca4*^{-/-}*Rdh8*^{-/-} and C57BL/6J mice. (A) Scotopic ERG. Stimulus luminance, 1 cd s/m². (B) Analysis of scotopic ERG with gradient stimulus luminance (n = 6). (C) Photopic ERG. Stimulus luminance, 10 cd s/m². (D) Analysis of photopic ERG with gradient stimulus luminance (n = 6). (E) c-wave ERG. Stimulus luminance, 30 cd s/m². (F) Analysis of c-wave ERG (n = 6). (G) Fundus imaging. Scale bars, 50 μm . (H) Assessments of RPE tight junctions. Scale bars, 20 μm . (I) Retinal thickness was analyzed by OCT (n = 6). Scale bars, 100 μm . (J) ONL thickness was analyzed by H&E staining (n = 6). Scale bars, 20 μm **p* < 0.05, ***p* < 0.01, ****p* < 0.001.

expression of senescence markers p16, p21 and γ H2AX, and in the protein levels of senescence-related signals p-mTOR and p-NF- κ B in the RPE/choroid of *Abca4*^{-/-}*Rdh8*^{-/-} mice at the age of 8 months (Fig. 5B and C). The qRT-PCR analysis revealed that the increase in mRNA levels of SASP-related genes *IL1b*, *IL6*, *MMP9*, *CCL2*, *CCL8* and *CXCL2* was significantly prevented by D + Q treatment in the RPE/choroid of 8-month-old *Abca4*^{-/-}*Rdh8*^{-/-} mice (Fig. 5D). These data suggest that long-term senolytic therapy with D + Q is effective in repressing RPE senescence and ameliorating retinal degeneration in *Abca4*^{-/-}*Rdh8*^{-/-} mice.

3.4. RPE degeneration in dry AMD patients with geographic atrophy is associated with lysosomal alkalization and oxidative stress

In dry AMD patients with geographic atrophy, significant fundus autofluorescence is observed [47], and this is closely related to lipofuscin accumulation caused by lysosomal dysfunction in the RPE [48]. We performed GSEA analysis on the RNA-seq data of the RPE from dry AMD patients with geographic atrophy compared to age-matched controls. The results revealed a negative correlation in genes for lysosome homeostasis (LYSOSOMAL_LUMEN_ACIDIFICATION, NES = -1.60 with

FDR q-value = 0.02; REGULATION_OF_LYSOSOMAL_LUMEN_PH, NES = -1.46 with FDR q-value = 0.05), suggesting that lysosomal alkalization was triggered (Fig. 6A). Lysosomal alkalization is closely associated with the dysfunction of the proton transporter V-ATPase complex [49]. Importantly, using GSEA, we identified a significantly negative correlation in genes for the V-ATPase complex of the RPE between dry AMD patients with geographic atrophy and age-matched controls (PROTON_TRANSPORTING_V_TYPE_ATPASE_COMPLEX, NES = -1.54 with FDR q-value = 0.03) (Fig. 6B). Meanwhile, a heatmap illustrated the expression patterns of V-ATPase complex-related genes *ATP6V1A*, *ATP6AP1*, *ATP6V0B*, *ATP6V0E2*, *ATP6V1E1*, and *ATP6V1D*, which were significantly downregulated in the RPE of dry AMD patients with geographic atrophy (Fig. 6C). Previous research shows that lipofuscin accumulation causes oxidative stress [24]. GSEA indicated that the RPE genes of dry AMD patients with geographic atrophy were enriched in oxidative stress-related gene sets (CELL_REDOX_HOMEOSTASIS, NES = -1.25 with FDR q-value = 0.16; OXIDATIVE_DAMAGE_RESPONSE, NES = 1.42 with FDR q-value = 0.05) (Fig. 6D and E). These findings disclose that the RPE of dry AMD patients with geographic atrophy displays oxidative stress, and V-ATPase complex dysfunction-mediated lysosomal alkalization.

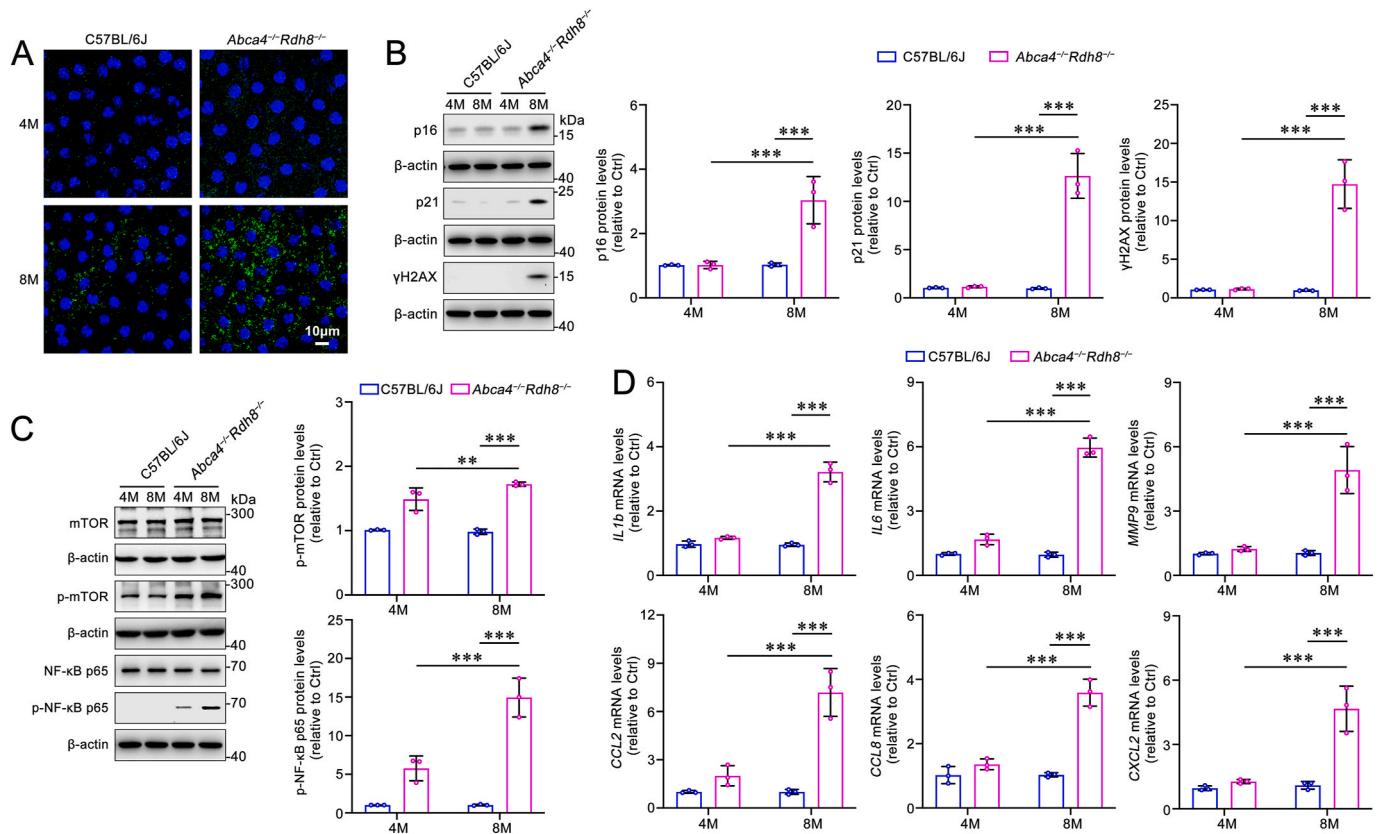


Fig. 3. *Abca4*^{-/-}*Rdh8*^{-/-} mice exhibit cellular senescence in the RPE with increasing age. (A) SA-β-gal staining for RPE/choroid flat mounts. Scale bars, 10 μm. (B) Protein levels of p16, p21 and γH2AX in the RPE/choroid were analyzed by immunoblotting (n = 3). (C) Protein levels of mTOR, p-mTOR, NF-κB p65 and p-NF-κB p65 in the RPE/choroid were evaluated using western blotting (n = 3). (D) The mRNA levels of *IL1b*, *IL6*, *MMP9*, *CCL2*, *CCL8* and *CXCL2* were examined by qRT-PCR (n = 3). ***p* < 0.01, ****p* < 0.001.

3.5. Long-term senolytic therapy with D + Q reduces lysosomal alkalization, lipofuscin accumulation and oxidative stress in the RPE of *Abca4*^{-/-}*Rdh8*^{-/-} mice

The qRT-PCR results showed that mRNA levels of V-ATPase complex-related genes, including *ATP6V1A*, *ATP6AP1*, *ATP6V0B*, *ATP6V0E2*, *ATP6V1E1*, and *ATP6V1D*, were clearly reduced in the RPE/choroid of *Abca4*^{-/-}*Rdh8*^{-/-} mice aged 8 months, with the most significant decrease observed in *ATP6V1A* compared to other genes (Fig. 7A). Prior studies have shown that V-ATPase complex-related genes are regulated by transcription factor EB (TFEB) [50]. Indeed, the results of immunoblotting demonstrated that protein levels of TFEB and *ATP6V1A* were remarkably decreased in the RPE/choroid of *Abca4*^{-/-}*Rdh8*^{-/-} mice at 8 months of age (Fig. 7B). The lysosomal pH was detected using the PDMPO probe [36]. The results showed that the lysosomal pH in the RPE of 8-month-old *Abca4*^{-/-}*Rdh8*^{-/-} mice was significantly increased compared to those in age-matched C57BL/6J mice (Fig. 7C), indicating that lysosomal alkalization occurred in the RPE. Given that lysosomal alkalization accelerates the accumulation of RPE lipofuscin [23,24], lipofuscin accumulation in the RPE was assessed using TEM and autofluorescence. The results showed that lipofuscin levels were visibly enhanced in the RPE of 8-month-old *Abca4*^{-/-}*Rdh8*^{-/-} mice compared to age-matched C57BL/6J mice (Fig. 7D and E). Moreover, it was found that the total antioxidant capacity, antioxidant GSH levels and antioxidant/pro-oxidant balance marker GSH/GSSG ratio were significantly reduced in the RPE/choroid of 8-month-old *Abca4*^{-/-}*Rdh8*^{-/-} mice (Fig. 7F and G), whereas the levels of protein carbonylation was remarkably increased (Fig. 7H). Immunoblotting showed that the expression of antioxidant-related enzymes catalase (CAT), SOD1 and GPX4 was decreased in the RPE/choroid of 8-month-old

Abca4^{-/-}*Rdh8*^{-/-} mice (Fig. 7I). Also, the levels of 4-HNE, an oxidative stress marker [51], was obviously increased in the RPE of 8-month-old *Abca4*^{-/-}*Rdh8*^{-/-} mice (Fig. 7I and J). These lines of evidence indicate that RPE degeneration in 8-month-old *Abca4*^{-/-}*Rdh8*^{-/-} mice involves lysosomal alkalization, lipofuscin accumulation and oxidative stress.

The qRT-PCR results showed that long-term senolytic therapy with D + Q obviously restored mRNA levels of V-ATPase complex-related genes (*ATP6V1A*, *ATP6AP1*, *ATP6V0B*, *ATP6V0E2*, *ATP6V1E1*, and *ATP6V1D*) in the RPE/choroid of 8-month-old *Abca4*^{-/-}*Rdh8*^{-/-} mice (Fig. 8A). D + Q treatment significantly increased protein levels of TFEB and *ATP6V1A* in the RPE/choroid of 8-month-old *Abca4*^{-/-}*Rdh8*^{-/-} mice (Fig. 8B), and it visibly decreased lysosomal pH in the RPE (Fig. 8C). Moreover, lipofuscin accumulation in the RPE of *Abca4*^{-/-}*Rdh8*^{-/-} mice aged 8 months was significantly prevented by long-term D + Q therapy (Fig. 8D and E). Long-term therapy with D + Q dramatically restored the total antioxidant capacity, GSH levels and GSH/GSSG ratio, and effectively inhibited the levels of protein carbonylation in the RPE/choroid of *Abca4*^{-/-}*Rdh8*^{-/-} mice at 8 months of age (Fig. 8F–H). In the RPE/choroid of 8-month-old *Abca4*^{-/-}*Rdh8*^{-/-} mice receiving long-term D + Q treatment, the expression of antioxidant-related enzymes CAT, SOD1 and GPX4 was significantly restored (Fig. 8I), and 4-HNE levels were obviously decreased (Fig. 8I and J). Overall, these findings imply that lysosomal alkalization, lipofuscin accumulation and oxidative stress in the RPE of *Abca4*^{-/-}*Rdh8*^{-/-} mice are capable of being ameliorated by long-term senolytic therapy with D + Q.

4. Discussion

Although research for therapy of dry AMD is currently booming,

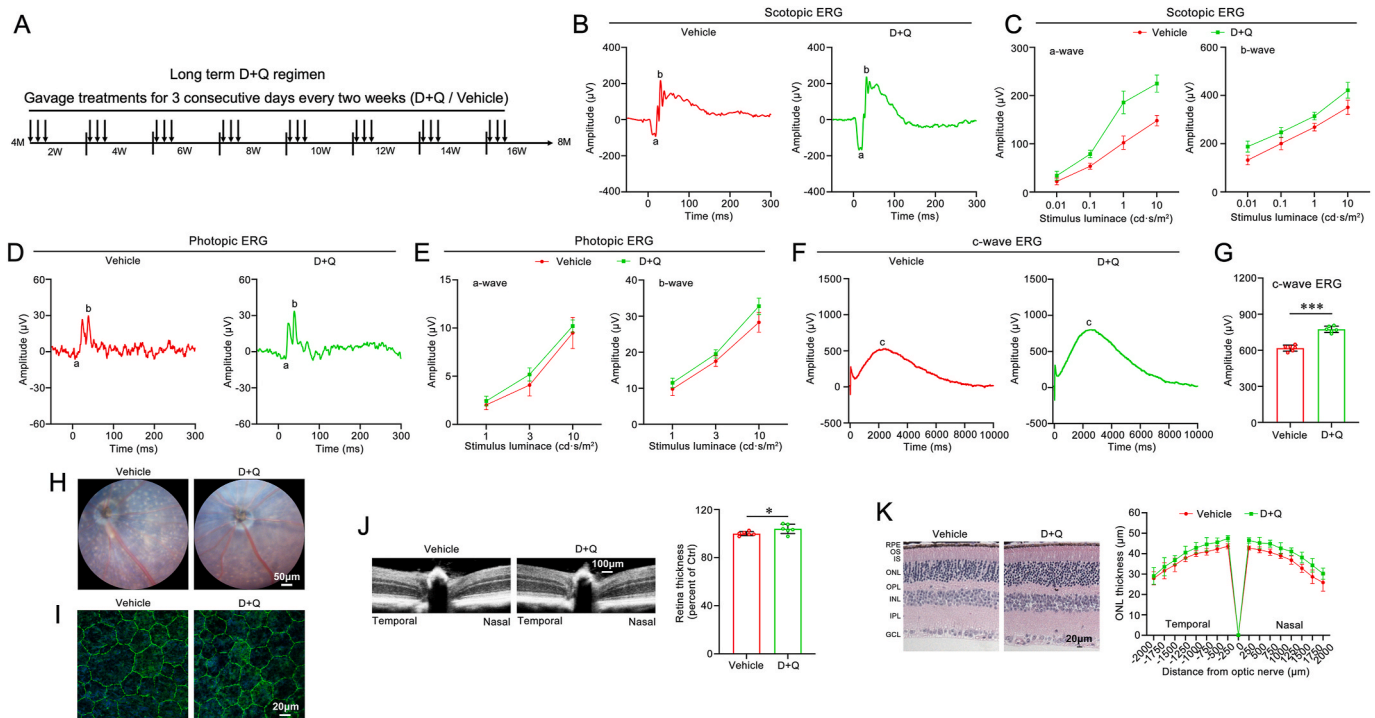


Fig. 4. Long-term senolytic therapy with D + Q effectively alleviates retinal degeneration in *Abca4*^{-/-}*Rdh8*^{-/-} mice. **(A)** Schematic diagram of long-term senolytic therapy with D + Q. **(B)** Scotopic ERG. Stimulus luminance, 1 cd s/m². **(C)** Analysis of scotopic ERG with gradient stimulus luminance (n = 6). **(D)** Photopic ERG. Stimulus luminance, 10 cd s/m². **(E)** Analysis of photopic ERG with gradient stimulus luminance (n = 6). **(F)** c-wave ERG. Stimulus luminance, 30 cd s/m². **(G)** Analysis of c-wave ERG (n = 6). **(H)** Fundus imaging. Scale bars, 50 μm. **(I)** Assessments of RPE tight junctions. Scale bars, 20 μm. **(J)** Retinal thickness was analyzed by OCT. Scale bars, 100 μm. **(K)** ONL thickness was evaluated by H&E staining (n = 6). Scale bars, 20 μm *p < 0.05, ***p < 0.001.

there are no effective interventions. Recent studies have demonstrated the importance of cellular senescence in retinal diseases [52,53]. In this study, we provided evidence for the involvement of RPE senescence in patients with dry AMD, and showed the efficacy of the long-term senolytic regimen with D + Q to delay the progression of age-dependent retinal degeneration in *Abca4*^{-/-}*Rdh8*^{-/-} mice, a mouse model of dry AMD. The long-term D + Q therapy indeed targeted cellular senescence and prevented lysosomal alkalization, lipofuscin deposition and oxidative stress in the RPE of *Abca4*^{-/-}*Rdh8*^{-/-} mice.

Retinal atrophy involves the senescence of retinal cells [52–54]. Specifically, the senescence of retinal ganglion cells promotes the progression of glaucoma [53], while the senescence of retinal vascular endothelium exacerbates neovascularization in diabetic retinopathy [52]. Here, our results showed that RPE senescence was positively correlated with geographic atrophy through analyzing RPE transcriptomic data from patients with dry AMD (Fig. 1). Compared with 8-month-old C57BL/6J mice, age-matched *Abca4*^{-/-}*Rdh8*^{-/-} mice exhibited retinal function decline and structural damage, among which the structural changes of the RPE were the most obvious (Fig. 2). Moreover, the RPE of *Abca4*^{-/-}*Rdh8*^{-/-} mice at 8 months of age displayed remarkably senescent phenotypes, as reflected by increased levels of SA-β-gal, p16, p21, γH2AX and SASP factors, and the activation of the mTOR/NF-κB signaling axis (Fig. 3). Taken together, these data imply that RPE senescence is closely associated with the progression of retinal degeneration in dry AMD.

Senolytic combination therapy, consisting of D and Q, synergistically targets and removes senescent cells. This regimen has potential therapeutic effects in alleviating age-related diseases, such as Alzheimer's disease and bone loss [55,56]. Past studies indicate that the elimination of senescent cells delays the progression of some retinal diseases, such as glaucoma and diabetic retinopathy [52,53]. Hyewon Chung and collaborators demonstrate that short-term oral treatment with ABT-263 (an inhibitor of Bcl-2 and Bcl-xL) selectively eliminates senescent RPE cells

and alleviates retinal degeneration in a Dox-induced mouse RPE senescence model [57]. Despite the established causal relationship between the senescence and retinal diseases, long-term senolytics therapy has not yet been employed in preclinical models of age-related dry AMD. In the current study, 4-month-old *Abca4*^{-/-}*Rdh8*^{-/-} mice were treated by gavage with D + Q or vehicle for 3 consecutive days every 2 weeks until they were 8 months old (Fig. 4A). It should be mentioned that, compared to C57BL/6J mice, retinal degeneration in *Abca4*^{-/-}*Rdh8*^{-/-} mice was not observed at 4 months of age, but was already significant at the age of 8 months (Fig. 2). *Abca4*^{-/-}*Rdh8*^{-/-} mice receiving D + Q treatment showed significant recovery of retinal function and structures, with the most obvious recovery in the RPE structures (Fig. 4). The long-term D + Q therapy decreased RPE senescence in 8-month-old *Abca4*^{-/-}*Rdh8*^{-/-} mice, as evidenced by the inhibition of the senescence-related mTOR/NF-κB signaling pathway and the decrease in the levels of senescence markers, including SA-β-gal, SASP factors, p16, p21, and γH2AX (Fig. 5).

Given that cellular senescence could be accompanied by lysosomal dysfunction [58], lysosome-related GSEA analysis were performed on the RNA-seq data of the RPE in dry AMD patients. As expected, the RPE of dry AMD patients exhibited obvious disorders of lysosomal acidification and a decline in V-ATPase complex expression (Fig. 6A and B). Likewise, the gene expression of the subunits that constitute the V-ATPase complex was significantly downregulated in the RPE/choroid of 8-month-old *Abca4*^{-/-}*Rdh8*^{-/-} mice (Fig. 7A). The subunits of the V-ATPase complex are mainly regulated by the mTOR/TFEB signaling pathway [59]. There is also evidence suggesting that lipofuscin accumulation in the RPE activates mTOR signaling [60]. The increase in protein levels of p-mTOR and the reduction in the expression of TFEB and ATP6V1A were clearly observed in the RPE/choroid of 8-month-old *Abca4*^{-/-}*Rdh8*^{-/-} mice (Figs. 3C and 7B).

Previous studies have revealed that lysosomal alkalization promotes the deposition of RPE lipofuscin [61]. In 8-month-old

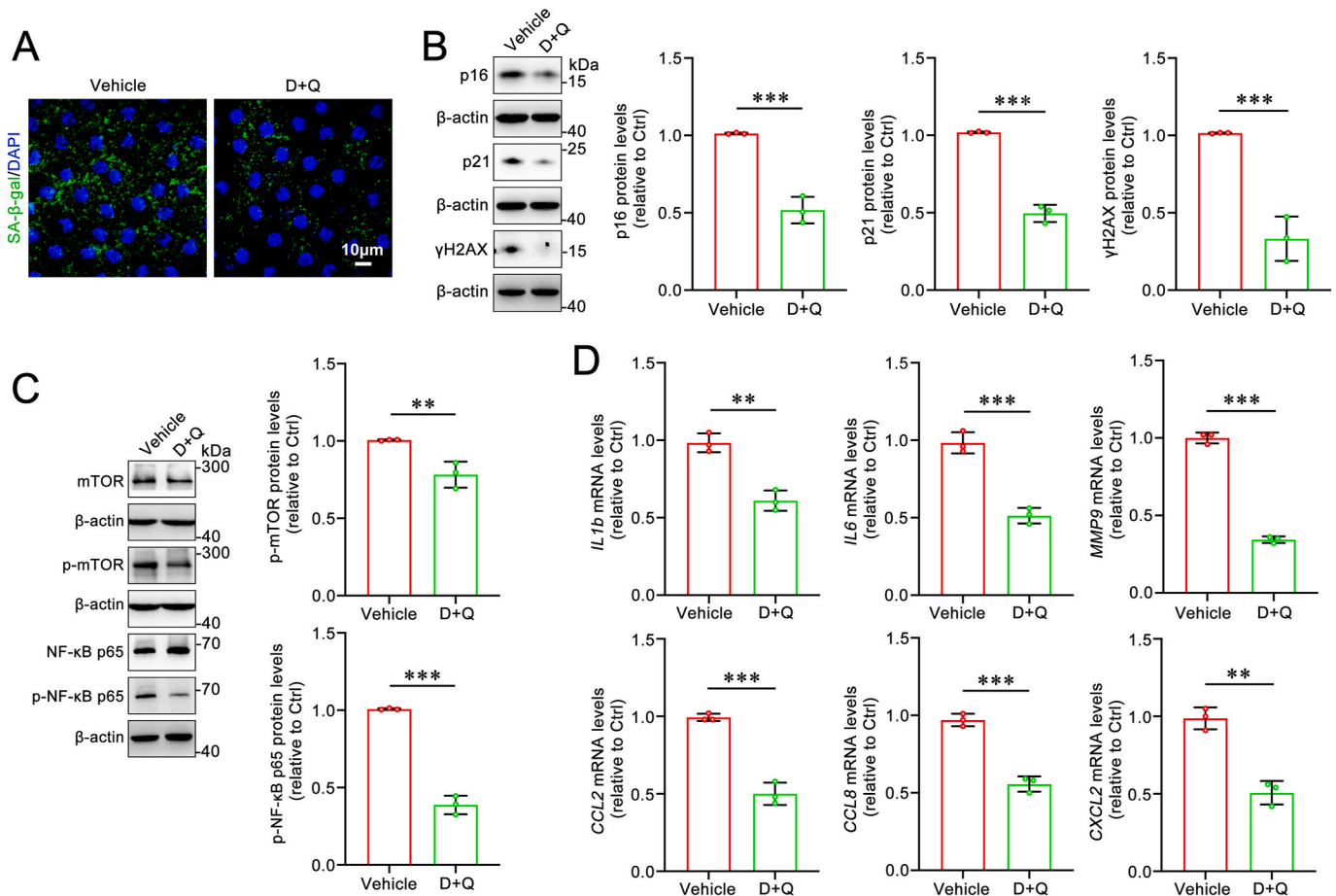


Fig. 5. Long-term senolytic therapy with D + Q decreases RPE senescence in 8-month-old *Abca4*^{-/-}*Rdh8*^{-/-} mice. **(A)** SA-β-gal staining of RPE/choroid flat mounts. Scale bars, 10 μm. **(B)** Protein levels of p16, p21 and γH2AX in the RPE/choroid were analyzed by immunoblotting (n = 3). **(C)** Protein levels of mTOR, p-mTOR, NF-κB p65 and p-NF-κB p65 in the RPE/choroid were evaluated using western blotting (n = 3). **(D)** The mRNA levels of *IL1b*, *IL6*, *MMP9*, *CCL2*, *CCL8* and *CXCL2* were examined by qRT-PCR (n = 3). ***p* < 0.01, ****p* < 0.001.

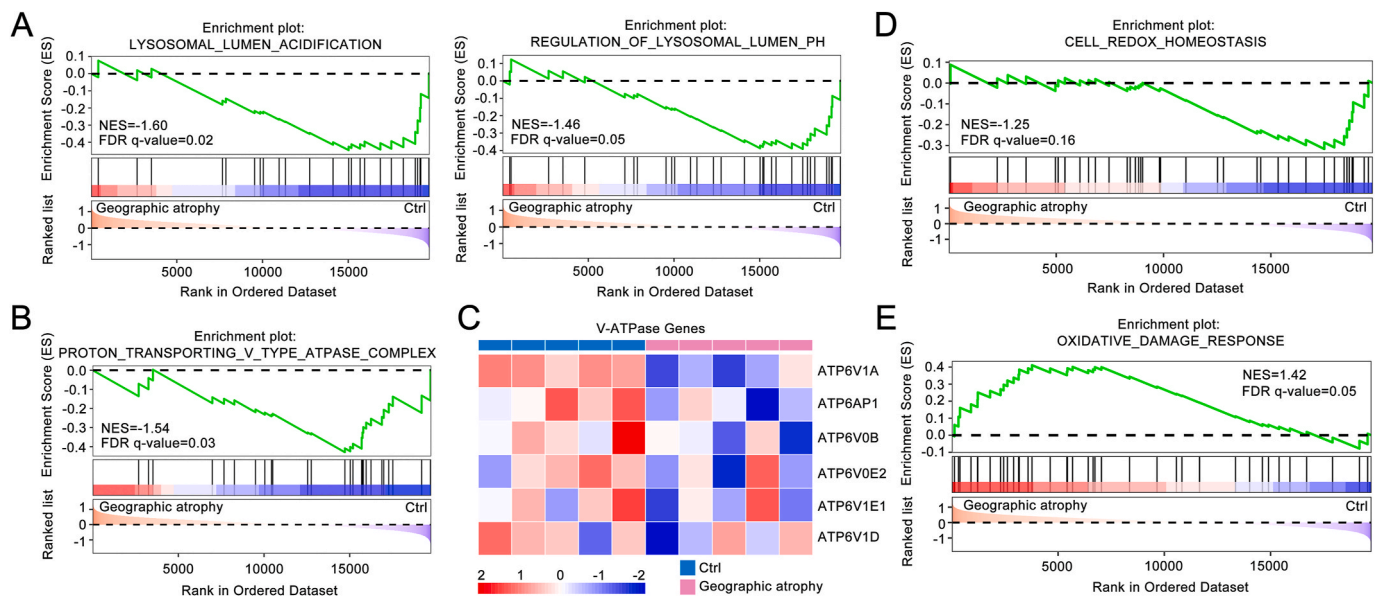


Fig. 6. The RPE of dry AMD patients with geographic atrophy shows lysosomal alkalization and oxidative stress. **(A, B)** GSEA was performed using gene sets 'LYSOSOMAL_LUMEN_ACIDIFICATION', 'REGULATION_OF_LYSOSOMAL_LUMEN_PH' and 'PROTON_TRANSPORTING_V_TYPE_ATPASE_COMPLEX', comparing RPE cells of dry AMD patients with geographic atrophy to normal control RPE cells. **(C)** A heatmap represents the profiles of V-ATPase genes in RPE cells of dry AMD patients with geographic atrophy, which were generated by comparison to the controls. **(D, E)** GSEA was carried out using gene sets 'CELL_REDOX_HOMEOSTASIS' and 'OXIDATIVE_DAMAGE_RESPONSE', comparing RPE cells of dry AMD patients with geographic atrophy to normal control RPE cells.

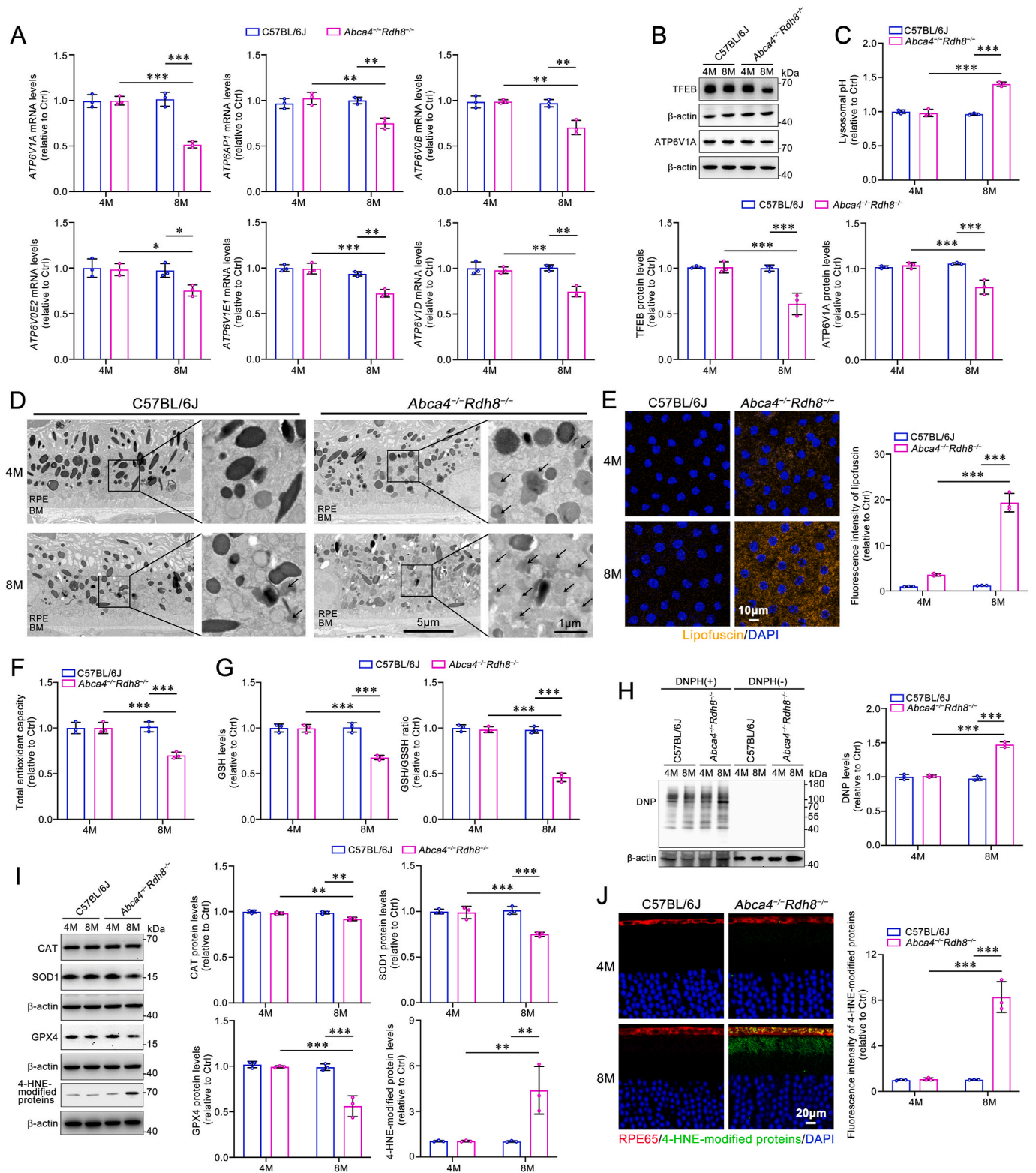


Fig. 7. *Abca4^{-/-}Rdh8^{-/-}* mice display lysosomal alkalization, lipofuscin accumulation and oxidative stress in the RPE with increasing age. (A) The mRNA levels of V-ATPase-related genes in the RPE/choroid were examined by qRT-PCR (n = 3). (B) The expression of TFEB and ATP6V1A in the RPE/choroid was analyzed using immunoblotting (n = 3). (C) Detection of lysosomal pH in the RPE (n = 3). (D) TEM images of lipofuscin in the RPE. Scale bars of low and high magnification images are 5 μ m and 1 μ m, respectively. (E) Autofluorescence analysis and quantification of RPE lipofuscin (n = 3). Scale bars, 10 μ m. (F) Detection of total antioxidant capacity (n = 3). (G) Detection of GSH levels and GSH/GSSG ratio (n = 3). (H) Detection of protein carbonyls (n = 3). (I) The levels of CAT, SOD1, GPX4 and 4-HNE-modified proteins in the RPE/choroid were determined by western blotting (n = 3). (J) Immunofluorescence of 4-HNE-modified proteins (n = 3). Scale bars, 20 μ m. * p < 0.05, ** p < 0.01, *** p < 0.001.

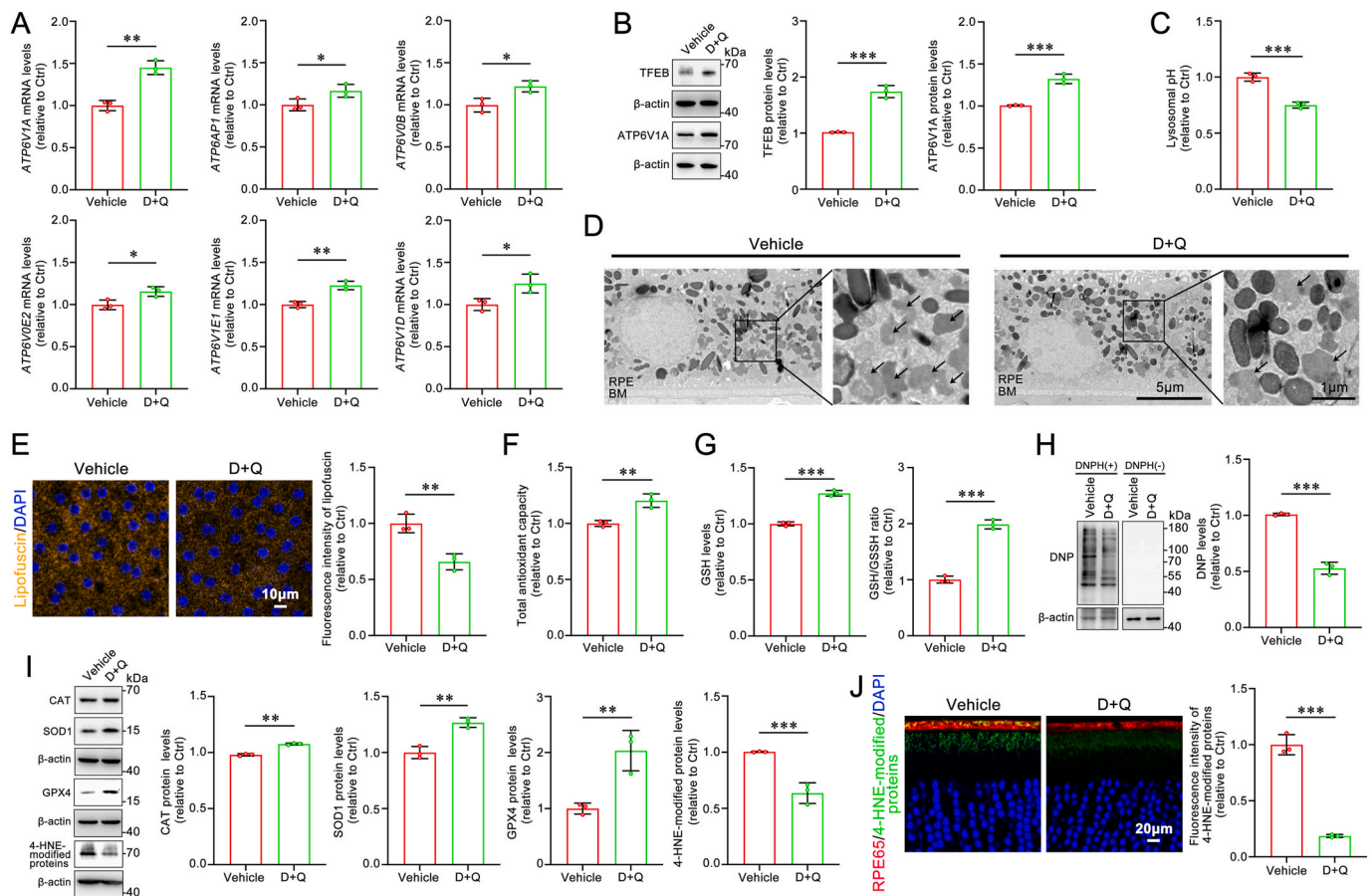


Fig. 8. Long-term senolytic therapy with D + Q mitigates lysosomal alkalization, lipofuscin accumulation and oxidative stress in the RPE of *Abca4*^{-/-}*Rdh8*^{-/-} mice. (A) The mRNA levels of V-ATPase-related genes in the RPE/choroid were examined by qRT-PCR (n = 3). (B) The expression of TFEB and ATP6V1A proteins in the RPE/choroid was analyzed by immunoblotting (n = 3). (C) Detection of lysosomal pH in the RPE (n = 3). (D) TEM images of lipofuscin in the RPE. Scale bars of low and high magnification images are 5 μ m and 1 μ m, respectively. (E) Autofluorescence analysis and quantification of RPE lipofuscin (n = 3). Scale bars, 10 μ m. (F) Detection of total antioxidant capacity (n = 3). (G) Detection of GSH levels and GSH/GSSG ratio (n = 3). (H) Detection of protein carbonyls (n = 3). (I) The levels of CAT, SOD1, GPX4 and 4-HNE-modified proteins in the RPE/choroid were determined by western blotting (n = 3). (J) Immunofluorescence of 4-HNE-modified proteins (n = 3). Scale bars, 20 μ m *p < 0.05, **p < 0.01, ***p < 0.001.

Abca4^{-/-}*Rdh8*^{-/-} mice, which are characterized by the accumulation of RPE lipofuscin [62], the lysosomal pH in the RPE was significantly increased (Fig. 7C). The accumulation of lipofuscin in the RPE is one of the important pathological features of dry AMD, which exhibits autofluorescence in the fundus of dry AMD patients [47]. In the present study, the autofluorescence resulting from lipofuscin deposition were distinctly visualized in the RPE of *Abca4*^{-/-}*Rdh8*^{-/-} mice aged 8 months (Fig. 7D and E).

Lipofuscin accumulation is reported to have the capacity to induce oxidative stress in the RPE [24]. GSEA indicated that the gene expression of the RPE in dry AMD patients was enriched in oxidative stress-related gene set (Fig. 6D and E). Compared with 8-month-old C57BL/6J mice, the RPE of age-matched *Abca4*^{-/-}*Rdh8*^{-/-} mice showed a significant decrease in the total antioxidant capacity, GSH levels, GSH/GSSG ratio, and the expression of antioxidant-related enzymes CAT, SOD1 and GPX4, and exhibited a remarkable increase in the content of protein carbonyls (Fig. 7F–I). The results of western blotting and immunofluorescence staining also verified that the levels of oxidative stress marker 4-HNE were visibly enhanced in the RPE of 8-month-old *Abca4*^{-/-}*Rdh8*^{-/-} mice (Fig. 7I and J). These findings suggest that RPE senescence contributing to dry AMD involves lysosomal alkalization, lipofuscin accumulation and oxidative stress. More importantly, we observed that long-term D + Q treatment effectively prevented retinal function decline and structural damage by alleviating cellular senescence, lysosomal alkalization, lipofuscin accumulation and

oxidative stress in the RPE of *Abca4*^{-/-}*Rdh8*^{-/-} mice at 8 months of age (Figs. 4, 5 and 8).

In summary, our data showed that long-term senolytic therapy with D + Q significantly alleviated retinal degeneration in a mouse dry AMD model through the repression of cellular senescence, lysosomal alkalization, lipofuscin accumulation, and oxidative stress in the RPE. A plausible mechanism for this discovery was depicted in Fig. 9. The accumulation of lipofuscin in the RPE triggers the phosphorylation of mTOR. On the one hand, p-mTOR drives cellular senescence by activating NF- κ B signaling to promote the expression of SASP factors, elevating the levels of senescence markers SA- β -gal, p16 and p21, and causing DNA damage. On the other hand, p-mTOR induces TFEB degradation, leading to a decrease in the expression of V-ATPase complex subunits and an increase in lysosomal pH. The accumulation of lipofuscin is further aggravated by lysosomal alkalization, which may elicit oxidative stress. Finally, RPE senescence and retinal atrophy occur. To our knowledge, the present study is the first to report the effect of long-term senolytic therapy with D + Q in dry AMD, and its underlying mechanism. This work not only deepens our understanding of the pathogenesis of dry AMD but also highlights the potential of long-term senolytic therapy as an intervention to the progression of dry AMD.

CRedit authorship contribution statement

Bo Yang: Conceptualization, Formal analysis, Investigation,

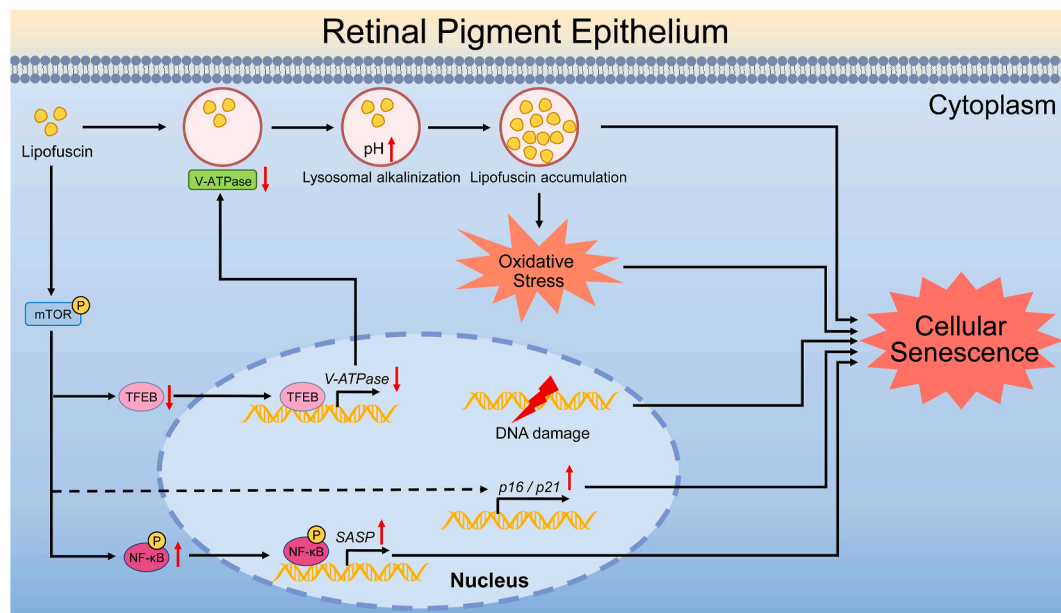


Fig. 9. Proposed mechanisms showing RPE senescence that contributes to dry AMD.

Methodology, Writing – original draft. **Kunhuan Yang**: Investigation. **Ruitong Xi**: Investigation. **Shiying Li**: Funding acquisition, Investigation, Resources. **Jingmeng Chen**: Resources. **Yalin Wu**: Conceptualization, Data curation, Formal analysis, Funding acquisition, Investigation, Methodology, Project administration, Resources, Software, Supervision, Validation, Visualization, Writing – original draft, Writing – review & editing.

Data availability

The data that support the findings of this study are obtained from the corresponding author upon reasonable request.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix. ASupplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.redox.2025.103716>.

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